
Preface

The generation of the *Semiconductor Industrial Hygiene Handbook* was a combined effort of the two primary authors with a significant contribution from Paul C. Manz. Paul provided the Continuous Gas Monitoring section of the Chemical Agents Chapter. This section provides a comprehensive review of the very important monitoring techniques for hazardous gases that are commonly used in the semiconductor industry. The primary authors are indebted for his contribution.

Information in the chapters on chemical and physical agents is based largely on published papers from periodicals that focus on the semiconductor industry. Dave Baldwin was a primary author on several of the key papers in this group. While some information is provided on hazard control, the main focus of these chapters is on hazard recognition.

The other primary chapter was generated by Mike Williams on ventilation. The use of massive amounts of air recirculation for contamination control in the fabrication areas can be both a help and a hindrance from an industrial hygiene perspective. This chapter details the use of ventilation for hazard control and its relationship with ventilation for contaminant control.

The scope of this book was expanded to include newer industrial hygiene concerns that are not specific to semiconductor industry: ergonomics, indoor air quality, personal protective equipment, plan review and records retention. While much of the information in these chapters can be applied to all industries, the focus and orientation is on how these issues are addressed in the semiconductor industry.

To provide additional detail on the processing technology, the silicon and III-V process descriptions on semiconductors were taken from the Cal/OSHA Semiconductor Industry Study. The original document was generated in 1981, with Mike Williams as the principal author. It was closely reviewed and additional new information was added in Appendices A–C. Additionally, a recent document on ion implanter hazards from a major implanter manufacturer was included as Appendix E. Lastly, a bibliography on processing technology for semiconductors was included as Appendix D for the reader who wants more in-depth process information.

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Lastly, the authors want to thank their respective families for the perseverance and patience for the longer than expected time it took to complete this task. For the Baldwin's: wife Susan Speciale, and their daughters Angela and Jennifer Baldwin; and for the Williams': wife Danelle, their son Brian, and daughter Jenny.

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Michael E. Williams
David G. Baldwin

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